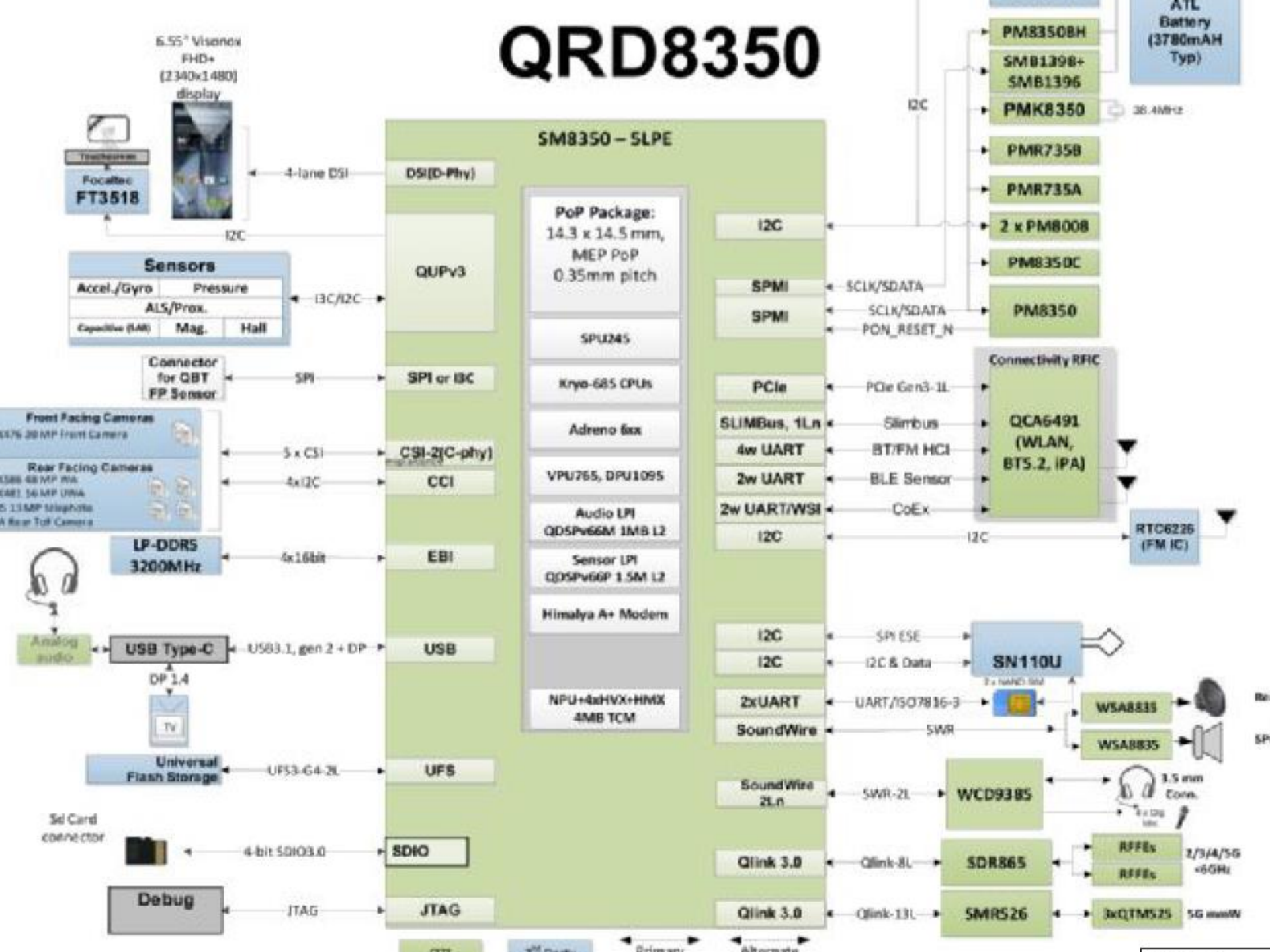


ZS590KS Sake Baseband Trouble shooting

Agenda

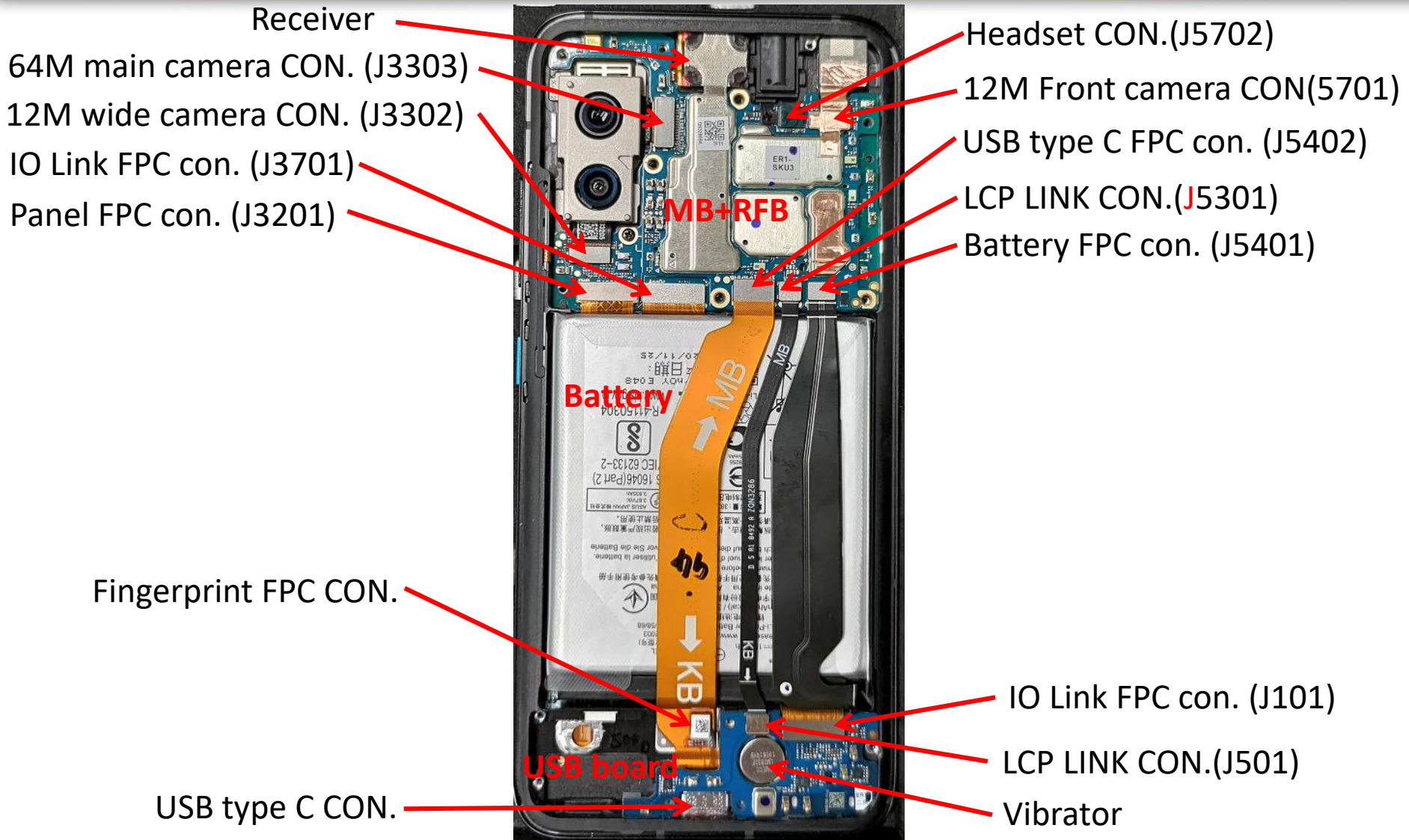
- System Block Diagram
- Phone Introduction
- PCB Introduction
 - Main-Board (MB)
 - RF-Board (RFB)
 - USB Sub-Board (USB SB)
 - MIC3 Sub-Board (MIC3 SB)
- Repair Guideline

QRD8350



Phone Introduction

Back side (1/2) – MB+RFB & USB SB



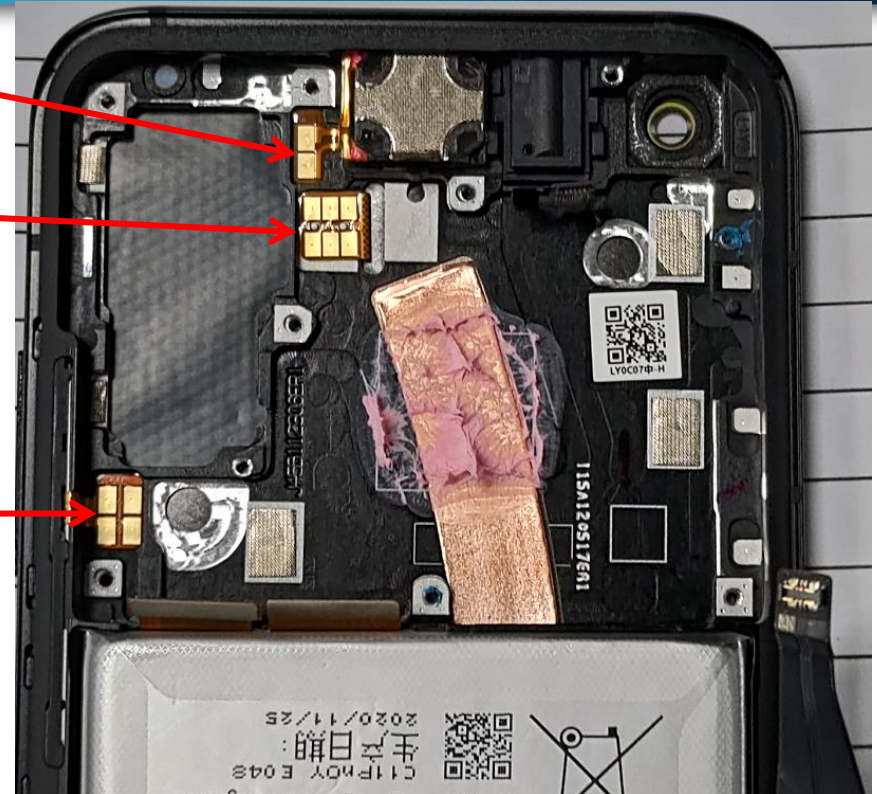
Phone Introduction

Back side (2/2) – MB+RFB & USB SB

Receiver contacted pads

P/L sensor CON. (J202)

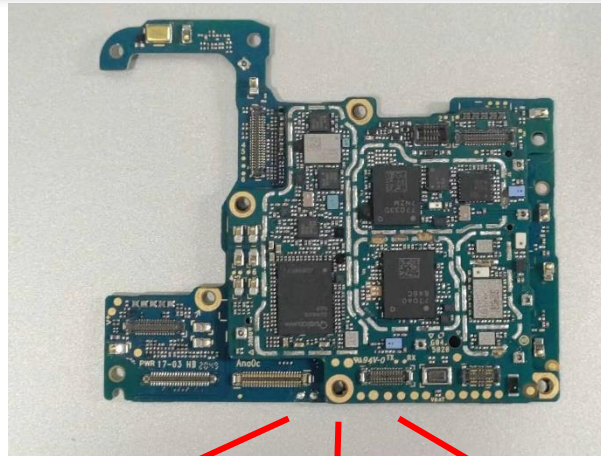
Side key contacted pads



Vibrator contacted pads



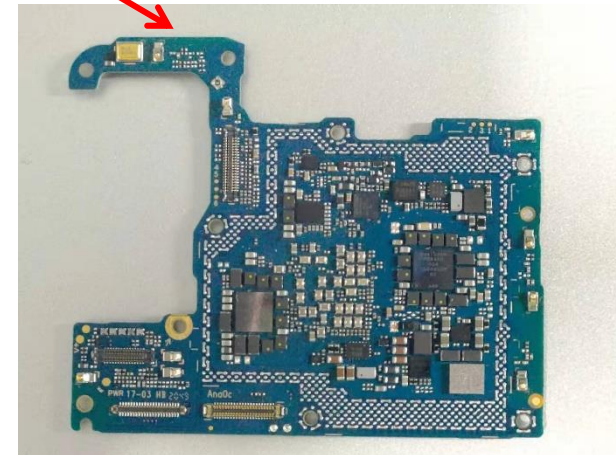
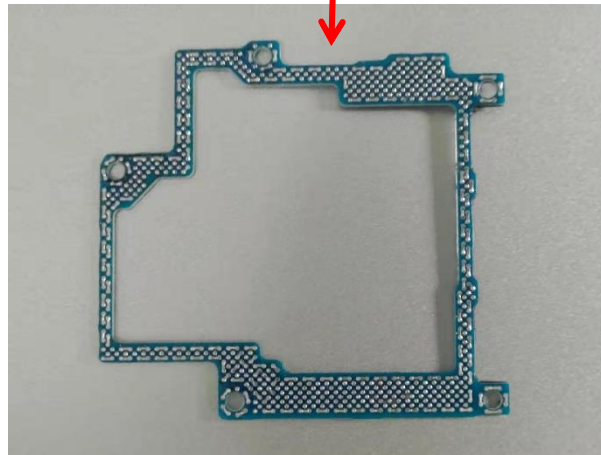
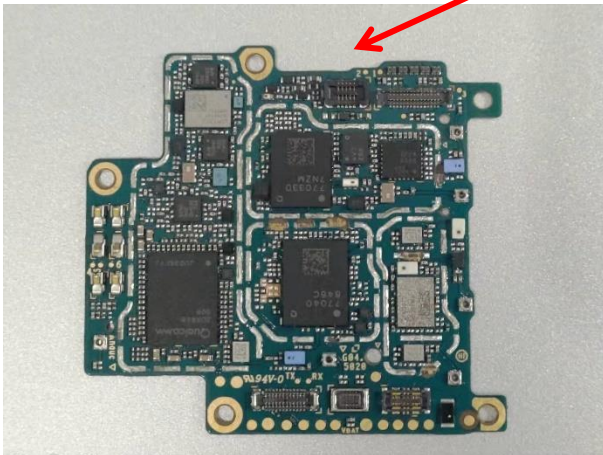
PCB Introduction



RFB

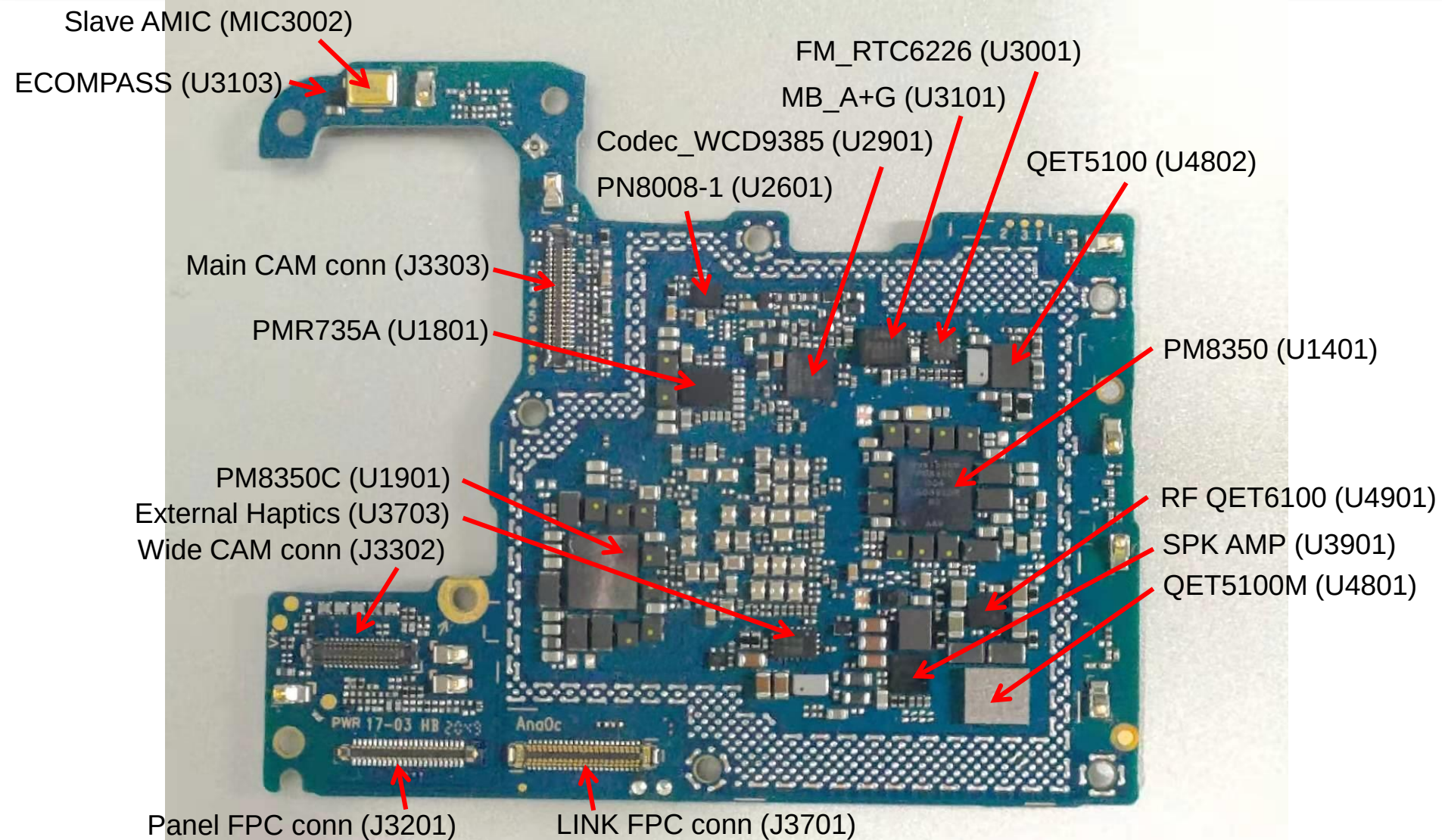
Interposer

MB



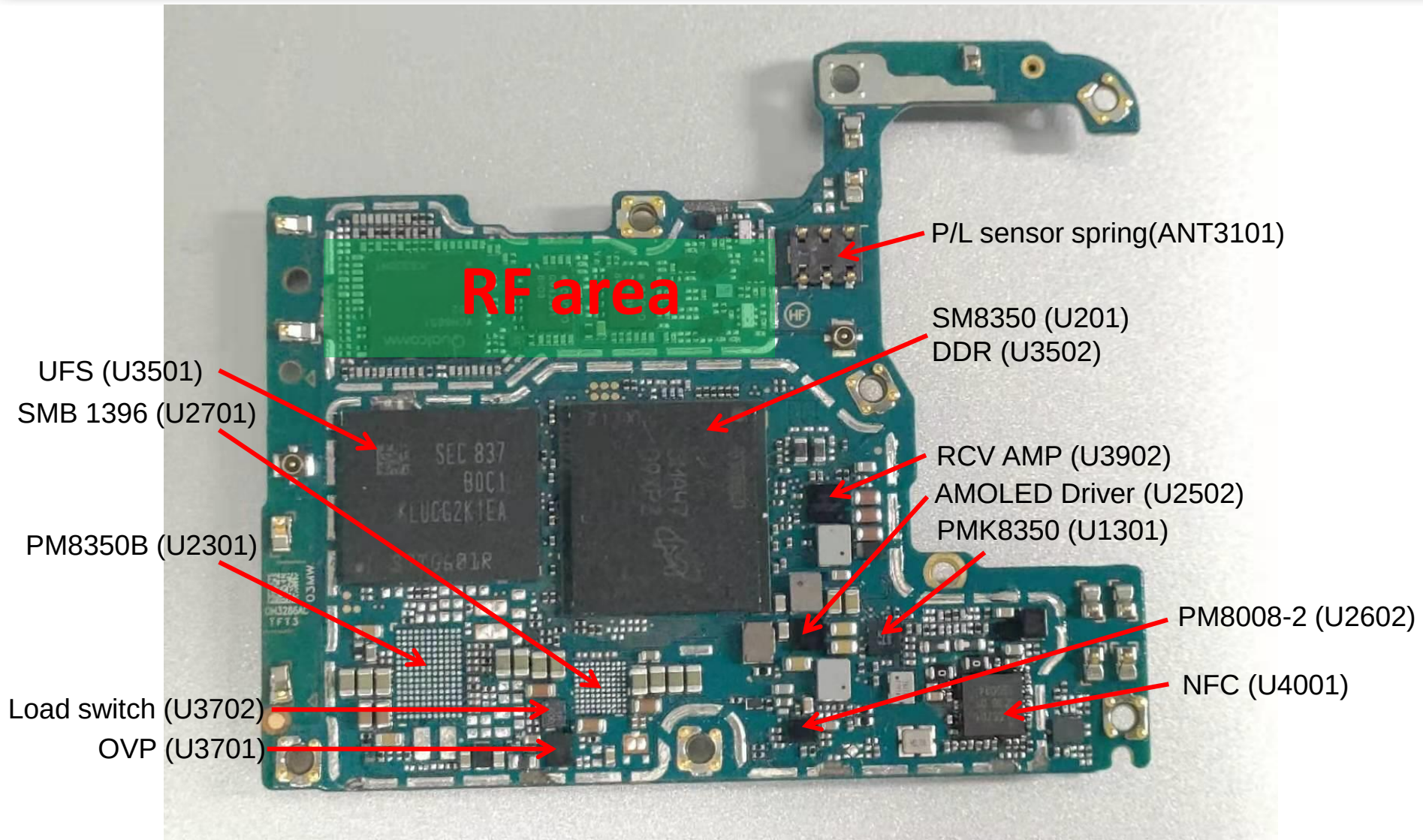
PCB Introduction

MB (1/2) - Bottom



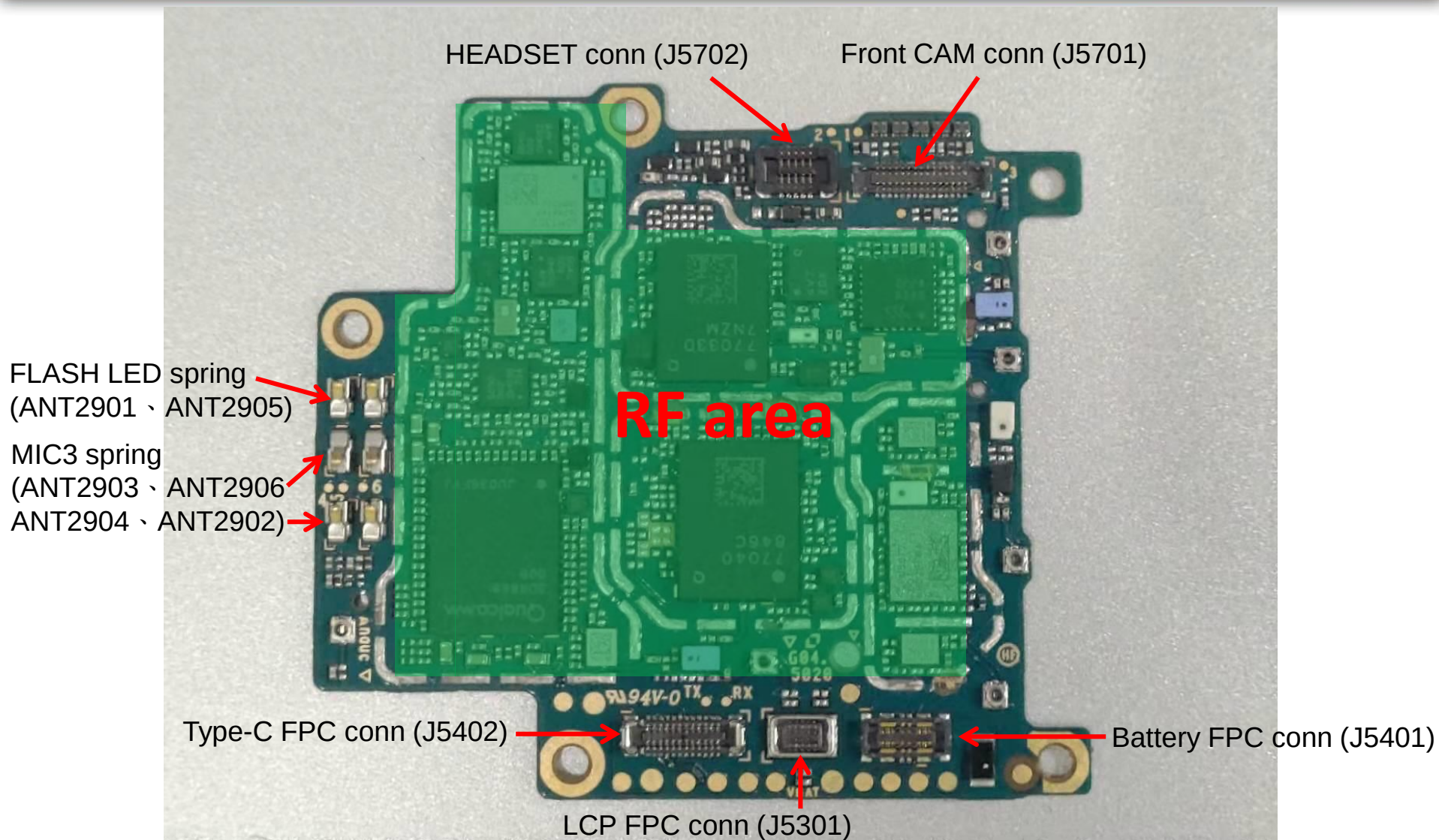
PCB Introduction

MB (2/2) - Top



PCB Introduction

RFB Bottom (1/1)



PCB Introduction

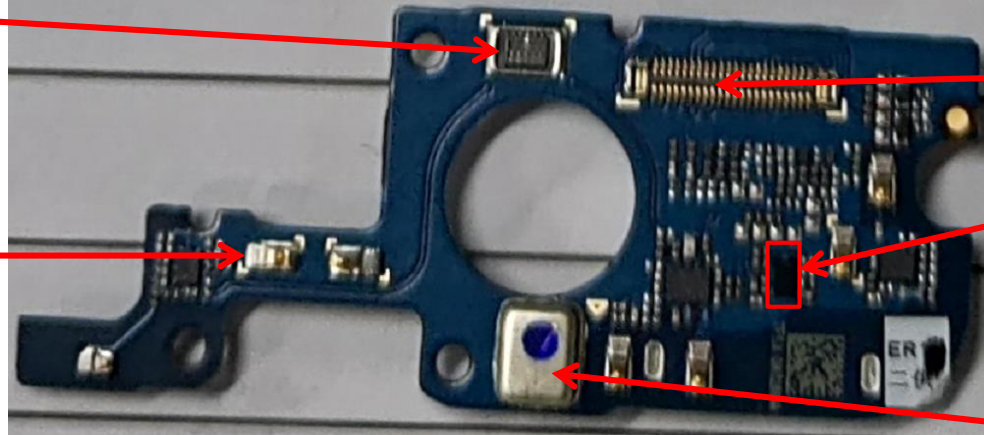
USB SB Top / Bottom (1/1)

LCP LINK CON.(J501)

IO Link FPC con. (J101)

SAR sensor (SAR401)

Main AMIC (MIC201)

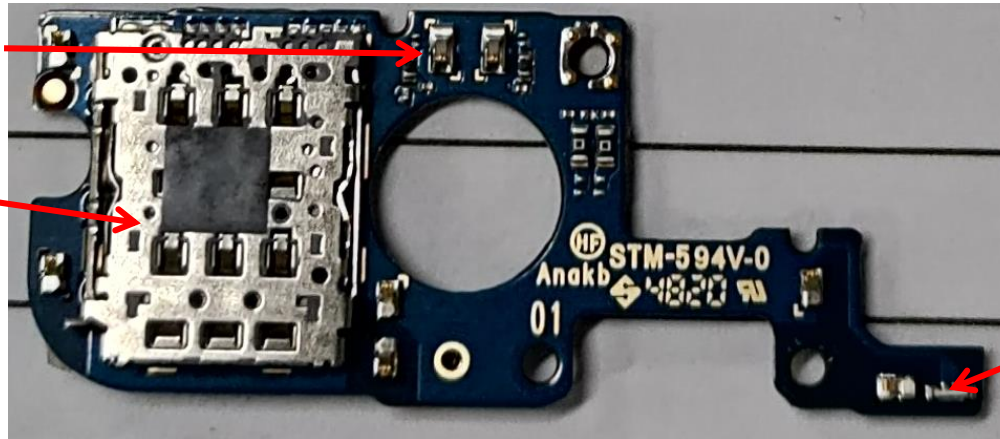


SPK spring (ANT201, ANT202)

Vibrator spring (ANT101, ANT102)

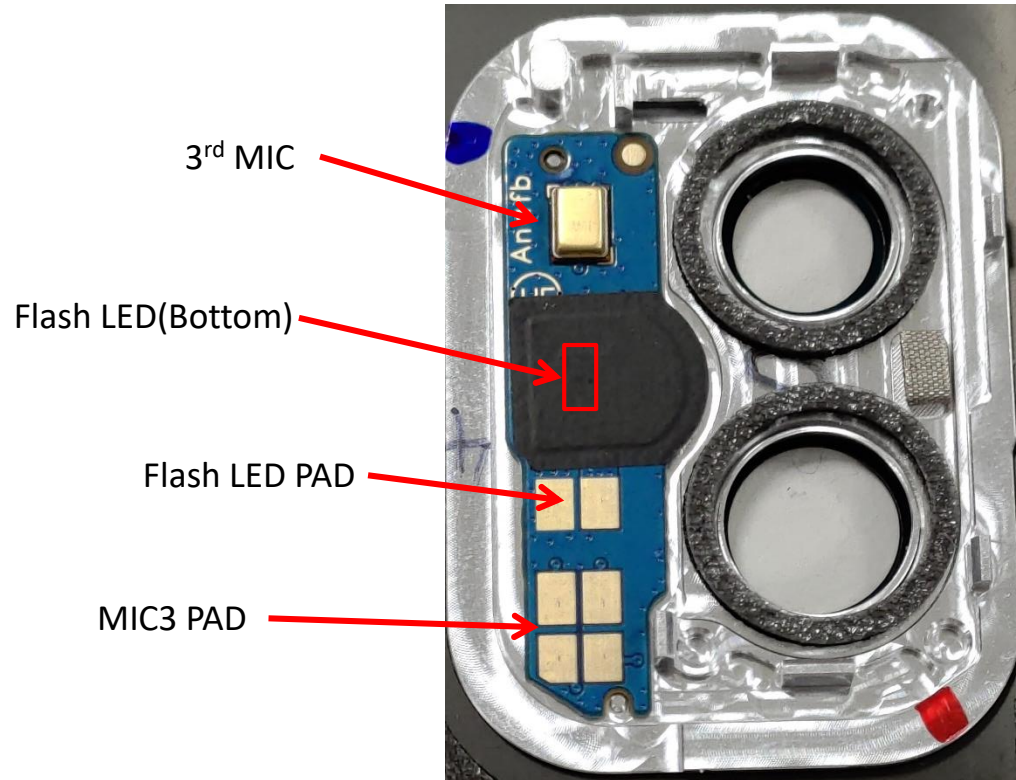
SIM CARD(J301, J302)

Indicator LED (LED201)



PCB Introduction

MIC3 SB Top / Bottom (1/1)



Repair Guideline

➤ Power on Fail

- UFS Fail
- Panel Fail
- Touch Fail
- 1st G+G Sensor Fail
- 2nd G+G Sensor Fail
- E-compass Sensor Fail
- Proximity Sensor Fail
- Fingerprint Fail
- 64M Main Camera Fail
- 12M Wide Camera Fail
- 12M Front Camera Fail

- USB Fail
- SIM Card Fail
- Vibrator Fail
- VOL Key Fail
- Indicator LED Fail
- Flash LED Fail
- Receiver Fail
- Speaker Fail
- Main Mic Fail
- Slave Mic Fail
- 3rd Mic Fail
- Headset Fail

Power ON Fail(1/3)

1. Check whether the battery voltage is correctly sent to the PCB or not.

→ Check the voltage of RFB TP3808.

1-1. No voltage: Check battery or RFB J5401 is normal.

1-2. Have voltage: Confirm the voltage is higher than 3.4V(boot voltage). If it's normal, then check the voltage of MB TP1304(SM_PS_HOLD) can keep 1.8V after press power key or plug USB cable.

1-2-1. MB TP1304 keep 1.8V: Check the panel or UFS is normal.

1-2-2. MB TP1304 cannot keep 1.8V, need to check voltage of VPH_PWR.

2. Voltage of VPH_PWR.

→ Check the impedance of MB L2401.2.

2-1. MB L2401.2 is short to ground.

Check the SMT status according to x-ray:

MB: PM8350(U1401), PM8350C(U1901), PM8350B(U2301), PM8008(U2601/U2602), EX_HAPTICS(U3703), SPK/RCV AMP(U3901/U3902), PMR735A(U1801), NFC(U4001), Interposer(U4101), QET5100M(U4801/U4802), QET6100M(U4901), QM42391TR13(U5705), QM45392TR13(U5809) and R, L, C components on the VPH_PWR path are normal.

RFB: QM77033D(U4501), QPM5677(U4701), RF5426(U5101), Interposer(U5401) and R, L, C components on the VPH_PWR path are normal.

2-2. The impedance of MB L2401.2 is normal, and is not short to ground.

Continued on next page Power ON Fail(2/3).

Power ON Fail(2/3)

VPH_PWR is close to battery voltage, but SM_PS_HOLD has no voltage.

-> 1. Check side key function is normal.

(1) Check side key FPC is normal.

(2) Check voltage of MB TP3839 is 0V, while pressing the Power key.

If it is not 0V, check MB ANT3603 and the contact to side key FPC is normal.

2. If side key is normal, then check power on sequence.

(1) Starting from VREG_BOB, measure the voltage of each power rail is normal or not.

If any power rail has a short circuit to ground, confirm the SMT status of all components on the path.

(2) If there is no short-circuit to ground, but one/multiple groups have no voltage or abnormal voltage, check the SMT status of all components on the power path

(3) If the status of all the component on this power rail are normal. Reheat or replace PMIC and SOC step by step.

(i) reheat PM8350B(U2301), (ii) reheat PM8350(U1401), (iii) reheat PM8350C(U1901), (iv) reheat SOC(U201)

(v) Replace PM8350B(U2301), (vi) Replace PM8350(U1401), (vii) Replace PM8350C(U1901)

(viii) Replace SOC(U201).

After the hardware boot process is completed, SM_PS_HOLD(TP1304) will be 1.8V. If there is still no display, there are two possibilities:

1. UFS Fail → Replace CPU/DDR/UFS.

2. OLED Fail → Check panel con(MB J3201) and panel is normal.

Power ON Fail(3/3)

Power on sequence

Table 3-20 Power-on sequence: LPDDR5 with no SD card

Event	Device name
VREG_COIN	PM8350
VREG_1P8_SYS	PM8350
PON trigger is asserted	
S1C	PM8350C
BOB	PM8350C
S11B	PM8350
S12B	PM8350
L10B	PM8350
S5B	PM8350
S9B	PM8350
L4B	PM8350
S2C	PM8350C
S6C_7C	PM8350C
S8C_9C	PM8350C
L8B	PM8350
S10B	PM8350
S10C	PM8350C
L6B	PM8350
VREF_MSM	PM8350
L5B	PM8350
L1B	PM8350
VREG_BB	PMK8350
L1C	PM8350C
L3B	PM8350

Table 3-20 Power-on sequence: LPDDR5 with no SD card

Event	Device name
S3C	PM8350C
L2B	PM8350
L7B	PM8350
L9B	PM8350
L6C	PM8350C
S1B	PM8350
LNBBCLK1	PMK8350
SLEEP_CLK	PMK8350

UFS Fail

1. Check the voltage of UFS

	Power rail	Net name	Test point	Voltage
UFS2.1	VCC	UFS2.1_3P0	TP3505	3.0
	VCCQ2	VREG_S10B_1P8	TP3507	1.8
UFS3.1	VCC	VREG_L7B_2P5	TP3505	2.5
	VCCQ1	VREG_L9B_1P2	TP3506	1.2

2. If the voltage is normal, reheat or replace the IC (UFS and CPU) step by step.

- (i) Reheat U3501 (UFS)
- (ii) Reheat U201 (CPU/DDR)
- (iii) Replace U3501
- (iv) Replace U201

Panel Fail

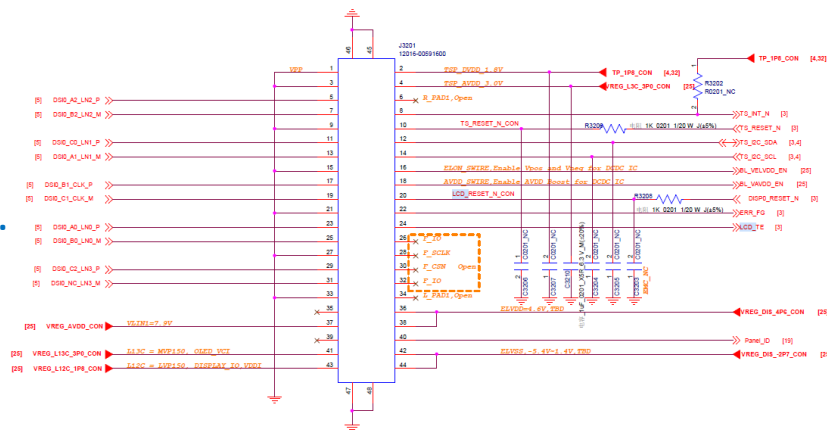
1. Check Panel first, then check SMT status of MB J3201.
2. Then check the power status.

VREG_AVDD_CON	C2540,R2508,C2539,C2543
VREG_DIS_4P6_CON	C2523,R2503,C2522,C2538,C2537
VREG_DIS_-2P7_CON	C2530,R2506,C2529,C2542,C2541
VREG_L12C_1P8_CON	C2533,R2507,C2532,C2531
VREG_L13C_3P0_CON	C2525,R2504,C2524

3. If the above steps are confirmed normal, but the voltages are still under the exception, reheat AMOLED driver(U2502) and PM8350C(U1901) sequentially.
4. If the problem is still not solved, replace AMOLED driver(U2502) and PM8350C(U1901) sequentially.

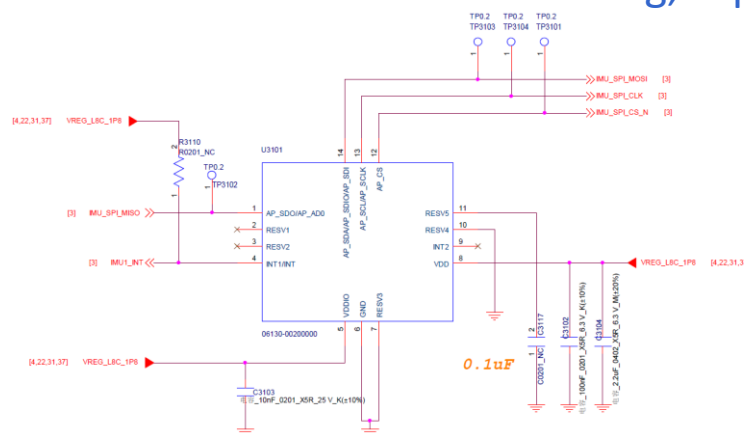
Touch Fail

1. Check Panel is normal.
2. Check SMT status of MB **J3201**.
3. Check the part of Panel Fail page.
4. Check TP_I2C_SDA: SMT status of MB R401,C3205.
5. Check TP_I2C_SCL: SMT status of MB R402,C3204.
6. Check TS_RESET_N: SMT status of MB R3206.
7. Check TS_INT_N: SMT status of MB R3202.
8. Check VREG_L3C_3P0_CON: SMT status of MB C2526, R2505, C2527, C3210.
9. Check TP_1P8_CON: SMT status of MB R3201, C2201, R3209, C3212, C3211, C3207.
10. If the above steps are confirmed normal, but the failure is still existing.
Reheat U1901 and U201 sequentially.
11. If the problem is still not solved, replace U1901 and U201 sequentially.



1st G+G Sensor Fail

1. Check the voltage of VREG_L8C_1P8 by measuring MB C3102.
2. If the voltage is abnormal, check VREG_L8C_1P8 connected capacitors first.
3. If the above components are normal, but VREG_L8C_1P8 is still 0 V. Check it's short to GND or open.
If it's short to GND, find the fail items.
If it's open, reheat U1901 first, then replace U1901.
4. If VREG_L8C_1P8 is normal, check U3101 by x-ray.
5. Measure the impedance or diode of IMU_SPI_CLK, IMU_SPI_MOSI, IMU_SPI_MISO, IMU_SPI_CS_N, IMU1_INT are normal or not.
6. If the above steps are done but the failure is still existing, replace U3101 and U201 sequentially.



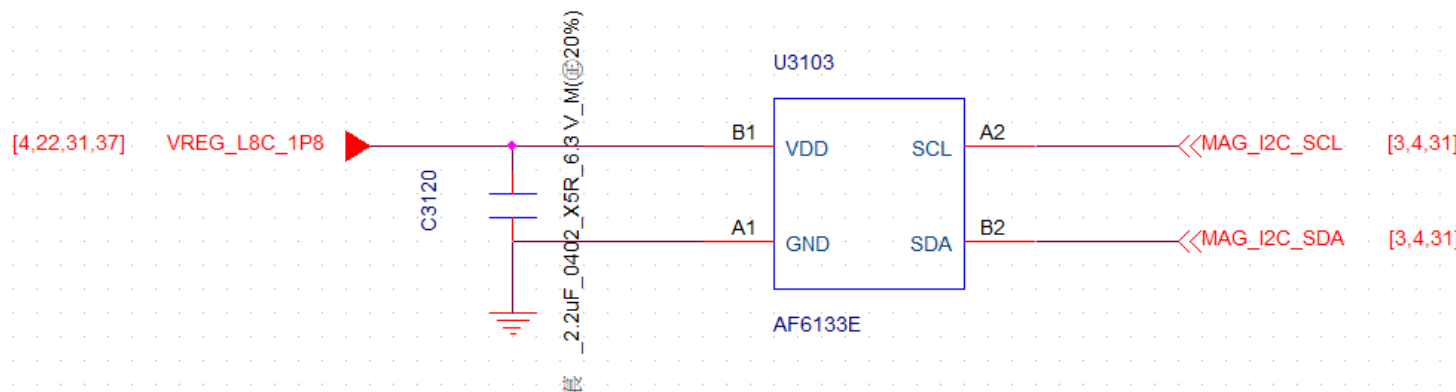
客供 ICM-40608

2nd G+G Sensor Fail

1. Check SMT status of MB J3303 and confirm the defect is with MB or camera module.
2. Defect is with camera module, replace camera module.
3. Defect is with MB, check SMT status of MB J3303, C3310, C3311, R3302, C2629, C3322, C3321, C0201, C2627.
4. Check the voltage of CAM012_DOVDD_1P8 by measuring MB C3322.
5. Check the voltage of CAM0_DRV_AVDD_2P9_MB by measuring MB R3302.
6. If CAM012_DOVDD_1P8, CAM0_DRV_AVDD_2P9_MB are normal
7. Measure the impedance or diode of GYRO_SDA_AP, GYRO_SCL_AP, GYRO_INT are normal or not.
8. If the above steps are done but the failure is still existing, replace MB U2602 and U201 sequentially.

E-compass Sensor Fail

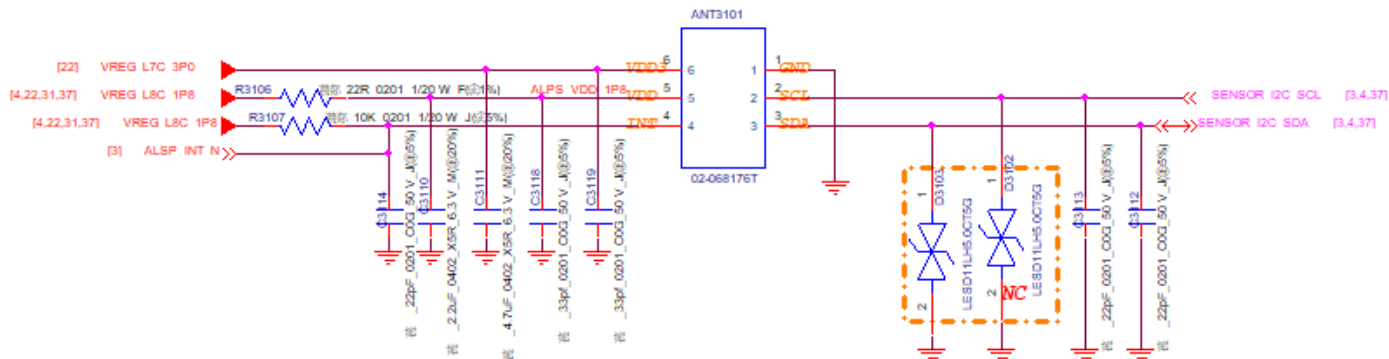
1. Check the voltage of VREG_L8C_1P8 by measuring MB C3120 ◦
2. If the voltage is abnormal, check VREG_L8C_1P8 connected capacitors first.
3. If the above components are normal, but VREG_L8C_1P8 is still 0 V. Check it's short to GND or open.
If it's short to GND, find the fail items.
If it's open, reheat U1901 first, then replace U1901.
4. If VREG_L8C_1P8 is normal, check U3103 by x-ray.
5. Measure the impedance or diode of MAG_I2C_SCL, MAG_I2C_SDA are normal or not.
6. If the above steps are done but the failure is still existing, replace U3103 and U201 sequentially.



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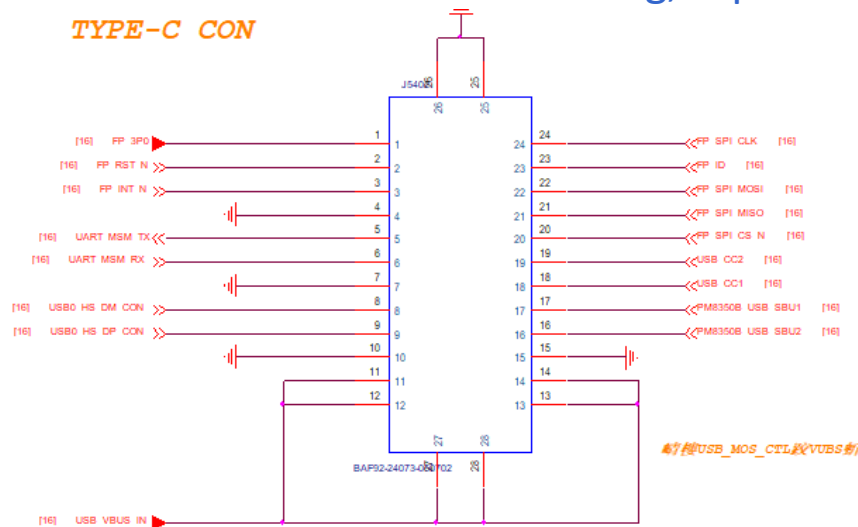
Proximity Sensor Fail

1. Check SMT status of MB ANT3101 and the contact with P/L sensor FPC.
2. Confirm the defect is with MB or P/L sensor FPC.
3. Defect is with P/L sensor FPC, replace P/L sensor FPC.
4. Defect is with MB, check the voltage of VREG_L8C_1P8 by measuring MB R3106. If it's abnormal, check the SMT status of MB R3106, C3110, C3118.
5. Check the voltage of VREG_L7C_3P0 by measuring MB C3111. If it's abnormal, check the SMT status of MB C3111, C3119.
6. Measure the impedance or diode of SENSOR_I2C_SCL, SENSOR_I2C_SDA, ALSP_INT_N are normal or not.
7. If the above steps are done but the failure is still existing, replace ANT3101 and U201 sequentially.



Fingerprint Fail

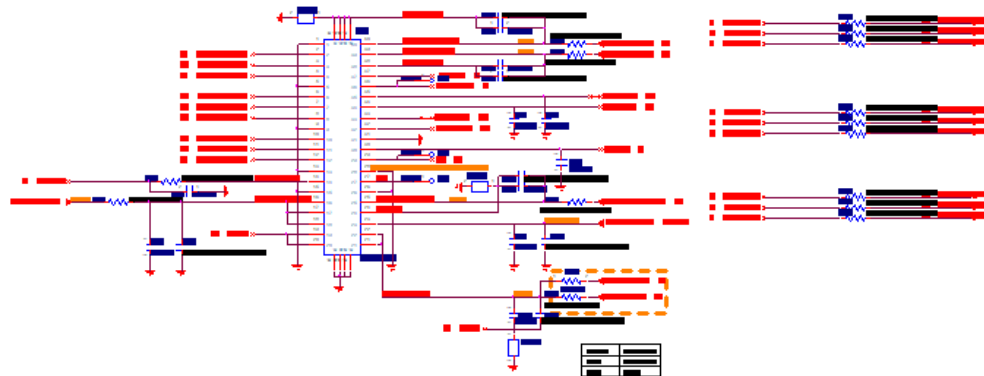
1. Check SMT status of RFB J5402.
2. Confirm the defect is with MB, Type-C FPC or Fingerprint FPC.
3. Defect is with Type-C FPC or Fingerprint FPC, replace Type-C FPC or Fingerprint FPC individually.
4. Defect is with MB, check the voltage of FP_3P0 by measuring RFB J5402. If it's abnormal, check the SMT status of interposer, MB U3202, C3209, C3208.
5. Measure the impedance or diode of FP_SPI_CLK, FP_ID, FP_SPI_MOSI, FP_SPI_MISO, FP_SPI_CS_N, FP_RST_N, FP_INT_N are normal or not.
6. If the above steps are done but the failure is still existing, replace RFB J5402 and MB U201 sequentially.



64M Main Camera Fail

1. Check SMT status of MB J3303.
2. Confirm the defect is with MB or 64M camera module.
3. Defect is with 64M camera module, replace 64M camera module.
4. Defect is with MB, check the voltage of CAM0_DVDD_1P1_MB, CAM0_AVDD_1P8_MB, CAM0_AVDD_2P9_MB, CAM0_DRV_AVDD_2P9_MB, CAM012_DOVDD_1P8, CAM0_VM_3P1_MB. If it's abnormal, check the SMT status of R, L, C components on these path.
5. Measure the impedance or diode of CSI0_A0, B0, C0, A1, B1, C1, A2, B2, C2, CCI_I2C0_SDA, SCL, CAM_MCLK0, CAM0_RESET_N are normal or not.
6. If the above steps are done but the failure is still existing, replace MB J3303 and MB U201 sequentially.

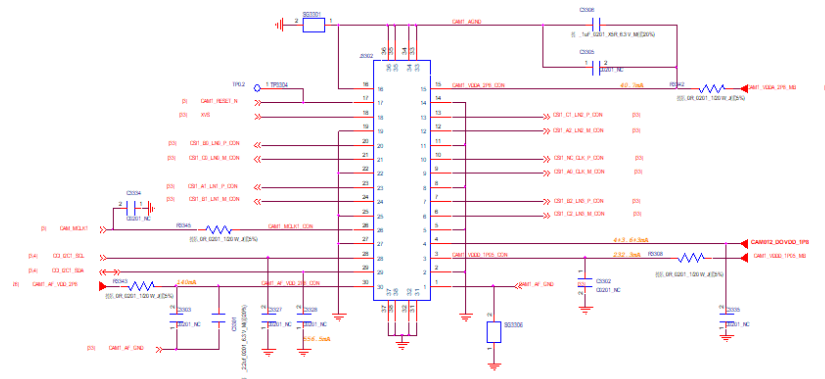
Main 64M



12M Wide Camera Fail

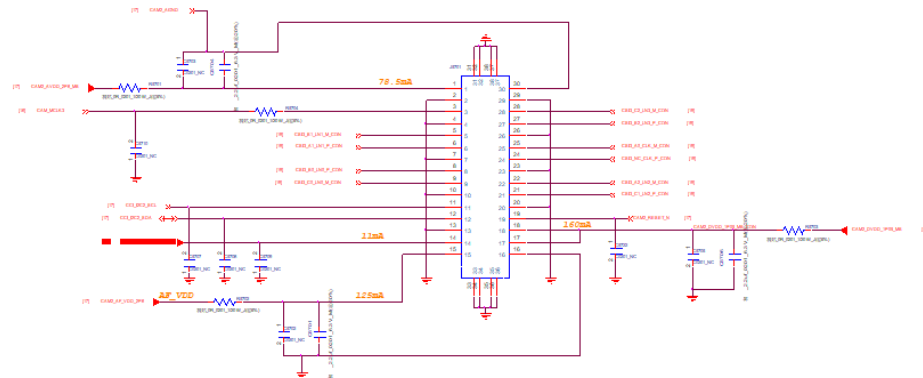
1. Check SMT status of MB J3302.
2. Confirm the defect is with MB or 12M wide camera module.
3. Defect is with 12M wide camera module, replace 12M wide camera module.
4. Defect is with MB, check the voltage of CAM1_VDDA_2P8_MB, CAM012_DOVDD_1P8, CAM1_VDDD_1P05_MB, CAM1_AF_VDD_2P8. If it's abnormal, check the SMT status of R, L, C components on these path.
5. Measure the impedance or diode of CSI1_A0_CLK_M/P, CSI1_B0_LN0_M/P, CSI1_B1_LN1_M/P, CSI1_A2_LN2_M/P, CSI1_C2_LN3_M/P, CCI_I2C1_SCL/SDA, CAM_MCLK1, CAM1_RESET_N are normal or not.
6. If the above steps are done but the failure is still existing, replace MB J3302 and MB U201 sequentially.

Wide 12M



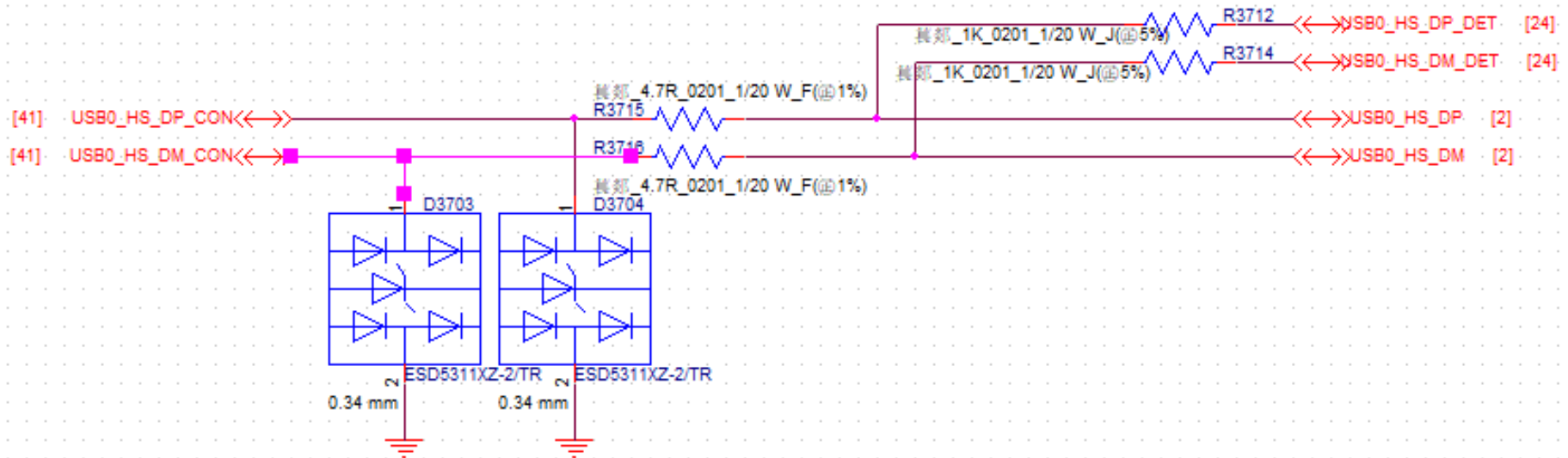
12M Front Camera Fail

1. Check SMT status of RFB J5701.
2. Confirm the defect is with MB or 12M front camera module.
3. Defect is with 12M front camera module, replace 12M front camera module.
4. Defect is with MB, check the voltage of CAM2_AVDD_2P8_MB, CAM012_DOVDD_1P8, CAM2_AF_VDD_2P8, CAM2_DVDD_1P05_MB. If it's abnormal, check the SMT status of R, L, C components on these path.
5. Measure the impedance or diode of CSI3_A0_CLK_M/P, CSI3_B0_LN0_M/P, CSI3_B1_LN1_M/P, CSI3_A2_LN2_M/P, CSI3_C2_LN3_M/P, CCI_I2C2_SCL/SDA, CAM_MCLK3, CAM2_RESET_N are normal or not.
6. If the above steps are done but the failure is still existing, replace RFB J5701, interposer and MB U201 sequentially. **Front 12M CAMERA**



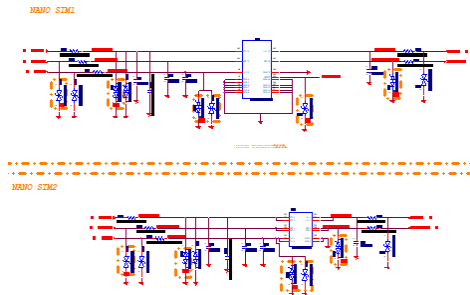
USB Fail

1. Check IO Link FC and SMT status of RFB J5402 is normal.
2. Confirm the defect is with MB or USB Type c FPC.
3. Defect is with MB, check SMT status of RFB J5402, interposer, MB D3703, D3704, R3715, R3716, R3712, R3714, U2301, U201 is normal.



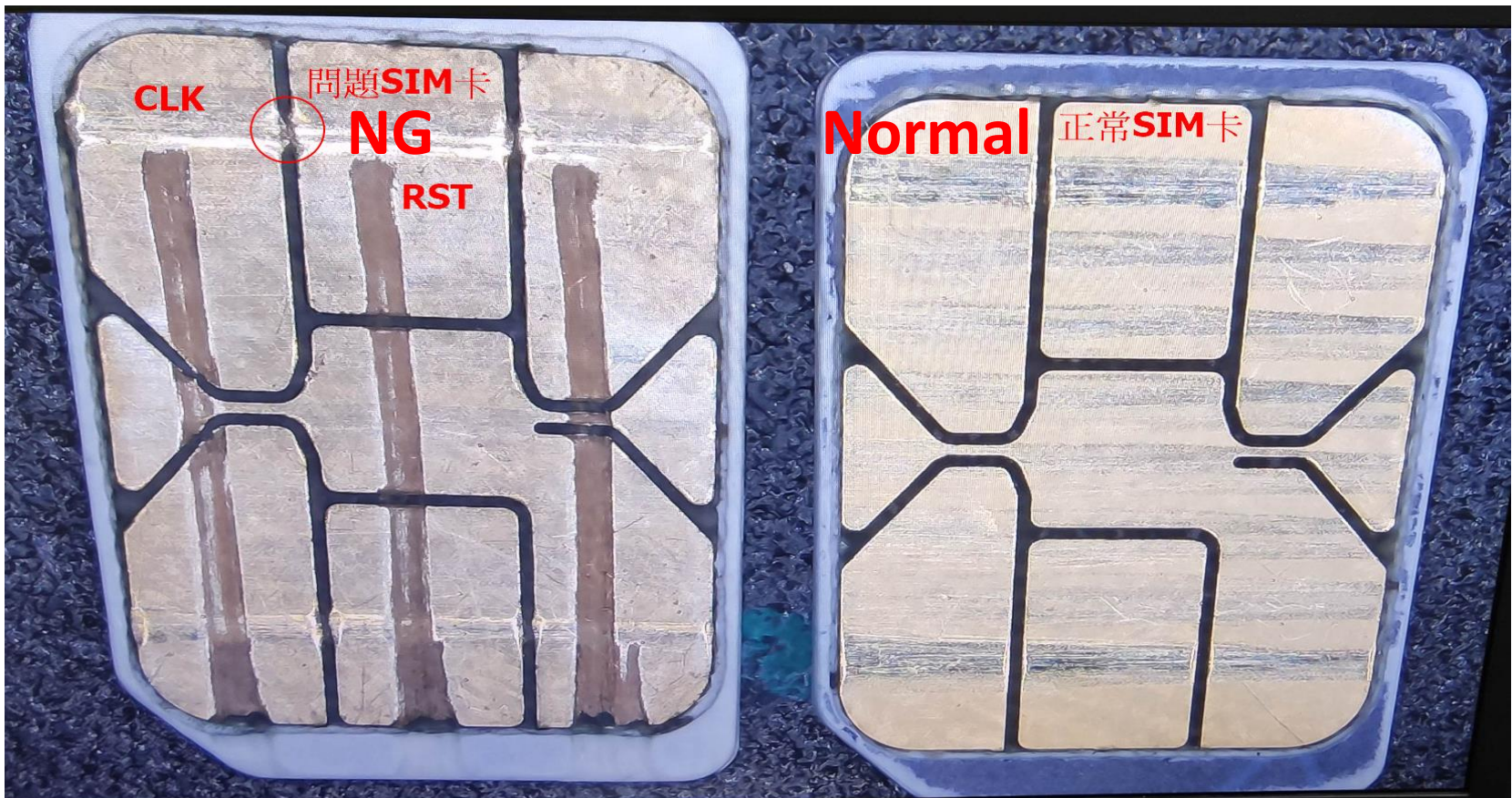
SIM Card Fail (1/3)

1. Check SIM tray, SIM1/2, IO Link FPC and SMT status of MB J3701, USB SB J101, J301, J302 is normal.
2. Confirm the defect is with IO Link FPC, USB SB or MB.
3. If SIM 1 is fail, check voltage of VCC_NFC1(USB SB R301) and the impedance or diode of UIM1_CLK(USB SB R305), UIM1_RESET(USB SB R302), UIM1_DATA(USB SB R303), NFC_SWIO_SIM1(USB SB R304) are normal or not.
4. If SIM 2 is fail, check voltage of VCC_NFC2(USB SB R306) and the impedance or diode of UIM2_CLK(USB SB R310), UIM2_RESET(USB SB R308), UIM2_DATA(USB SB R307), NFC_SWIO_SIM2(USB SB R309) are normal or not.
5. Check voltage of UIM_PRESENT(USB SB D309) will up to 1.8V after SIM tray insert. If it is always 0V, Check SMT status of USB SB J101, J301, D309 and MB J3701, R3005, R3006.
6. If the above steps are done but the failure is still existing, check the SIM card or replace MB U201 sequentially.



SIM Card Fail (2/3)

Confirm there is a short circuit or abnormality in the appearance of the SIM card



SIM Card Fail (3/3)

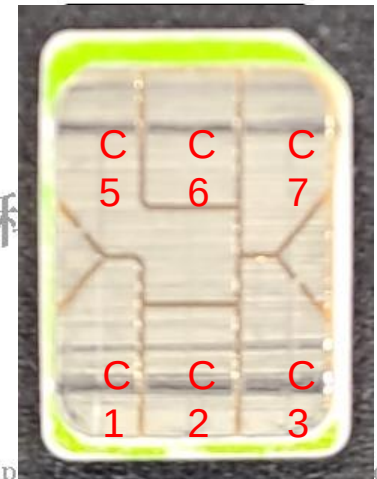
Measure the impedance between each pin

The following are the reference values for measuring China Mobile's SIM card

	Positive probe(+)						
		C1	C2	C3	C5	C6	C7
Negative probe(-)	C1		6.4M	6.4M	1.5M	open	60.7k
	C2	9.3M		33M	1.9M	open	9.3M
	C3	13.2M	38.7M		2M	open	13.3M
	C5	7.2M	21.8M	22.1M		open	7.3M
	C6	open	open	open	open		open
	C7	60.6K	6.5M	6.5M	1.6M	open	

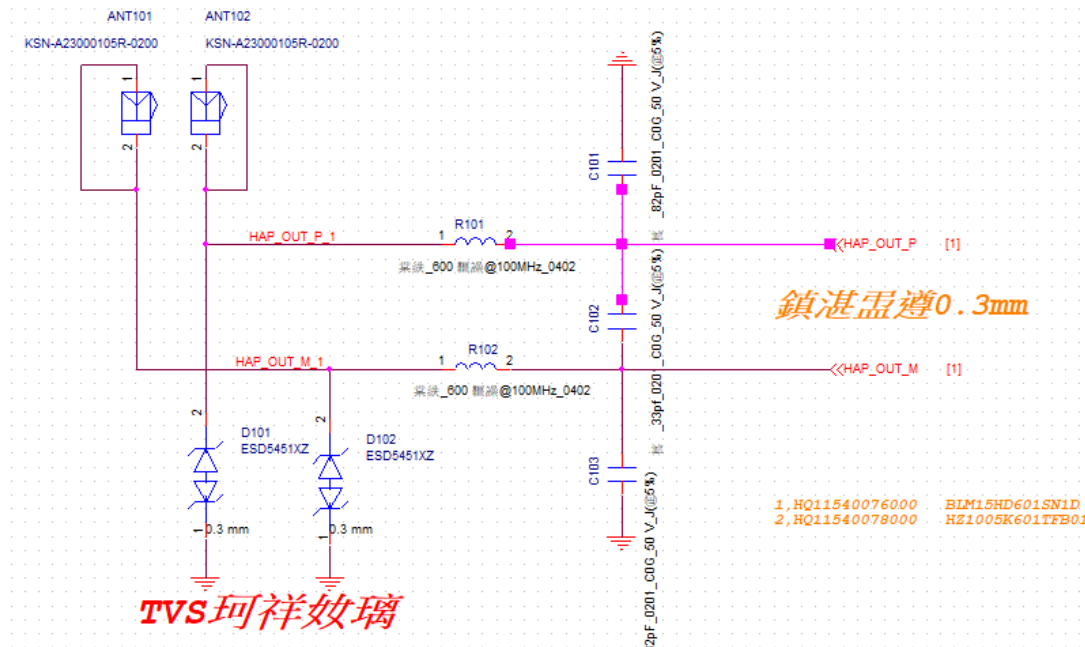
【 SIM Card Pin Assignments 】

PIN	Name
C1	Vcc
C2	RST
C3	CLK
C4	Reserved
C5	GND
C6	Vpp
C7	I/O
C8	Reserved



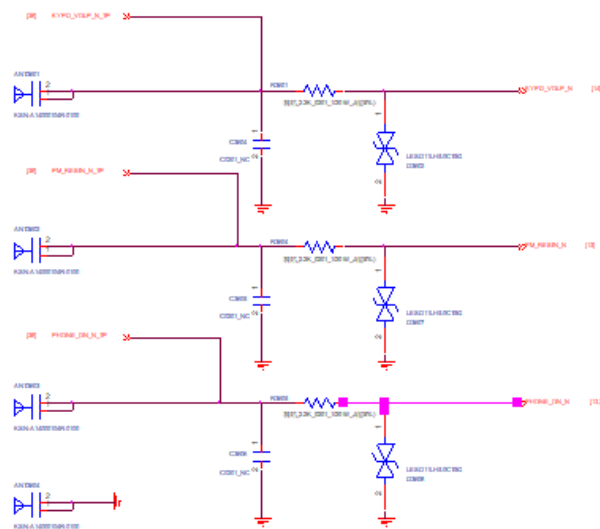
Vibrator Fail

1. Check Vibrator, IO Link FC and SMT status of USB SB ANT101, ANT102 is normal.
2. Check impedance of Vibrator is not short.
3. Confirm the defect is with Vibrator, IO Link FPC, USB SB or MB.
4. Defect is with Vibrator or IO Link FPC, replace it individually.
5. Defect is with USB SB, check SMT status of USB SB ANT101, ANT102, R101, R102, C101, C102, C103, D101, D102, J101 is normal.
6. Defect is with MB, check SMT status of MB R3719, R3720, U3703, U201 is normal.



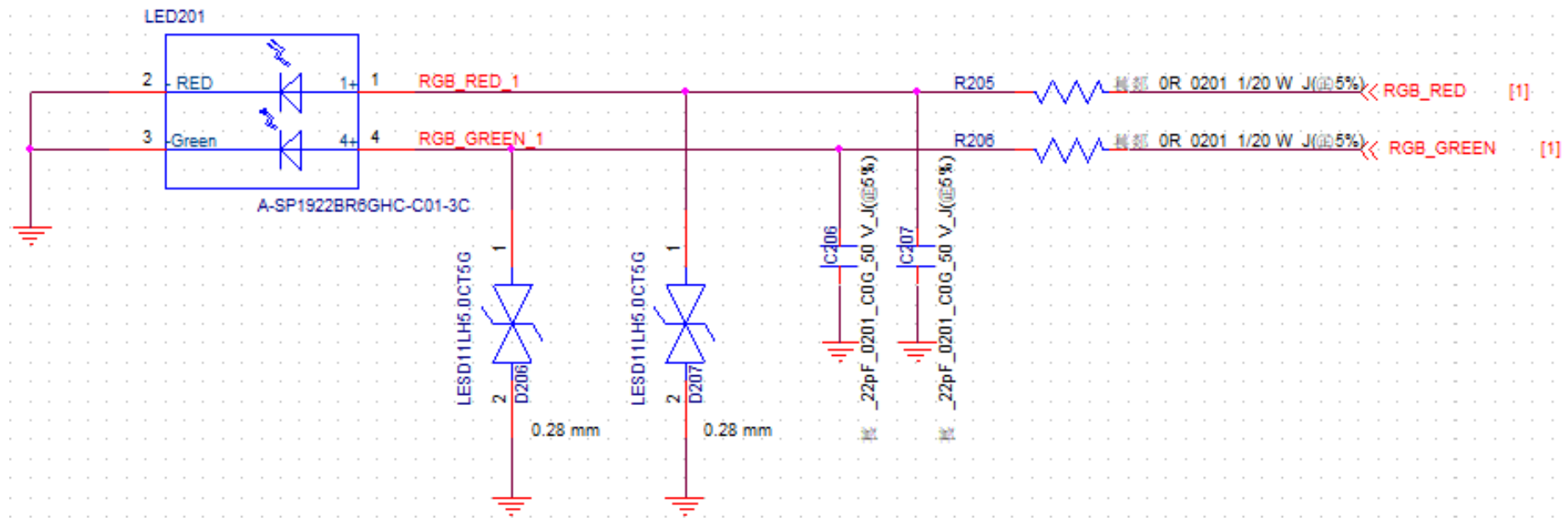
VOL Key Fail

1. Check side key FPC and the contact to MB ANT3601, ANT3602, ANT3603, ANT3604 is normal.
2. Confirm the defect is with side key FPC or MB.
3. Check voltage of KYPD_VOLP_N(R3601) is from 1.8V to 0V, while pressing volume up key. If not, check SMT status of MB R3601, C3604, D3602 and U1401.
4. Check voltage of PM_RESIN_N(R3604) is from 1.8V to 0V, while pressing volume down key. If not, check SMT status of MB R3604, C3605, D3607 and U1301.
5. Check voltage of PHONE_ON_N(R3605) is from 1.8V to 0V, while pressing power key. If not, check SMT status of MB R3605, C3606, D3608, U1301, U2301.



Indicator LED Fail

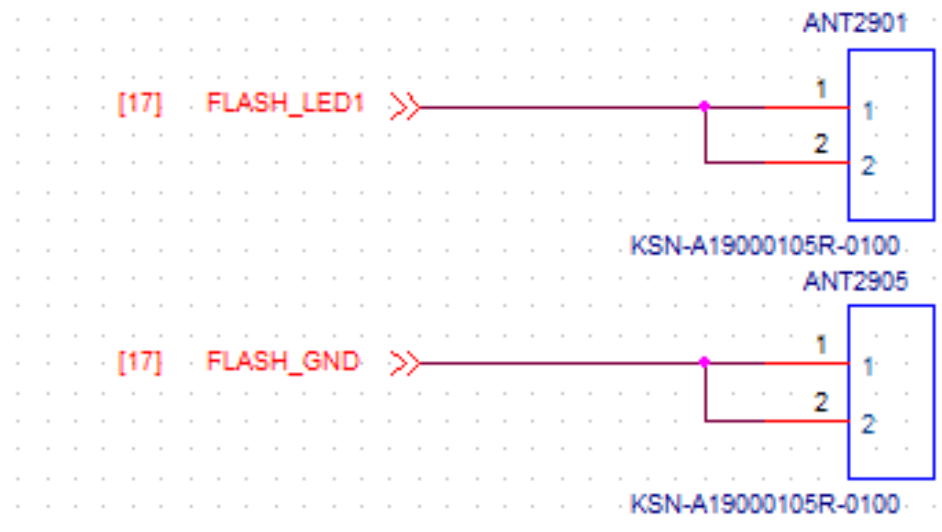
1. Check IO Link FPC, SMT status of MB J3701, USB SB J101, LED201 is normal.
2. Confirm the defect is with IO Link FPC, USB SB or MB.
3. Defect is with USB SB, check SMT status of USB SB J101, LED201, R205, R206, C206, C207, D206, D 207 is normal.
4. Defect is with MB, check SMT status of MB J3701, U1901 is normal.



Flash LED Fail

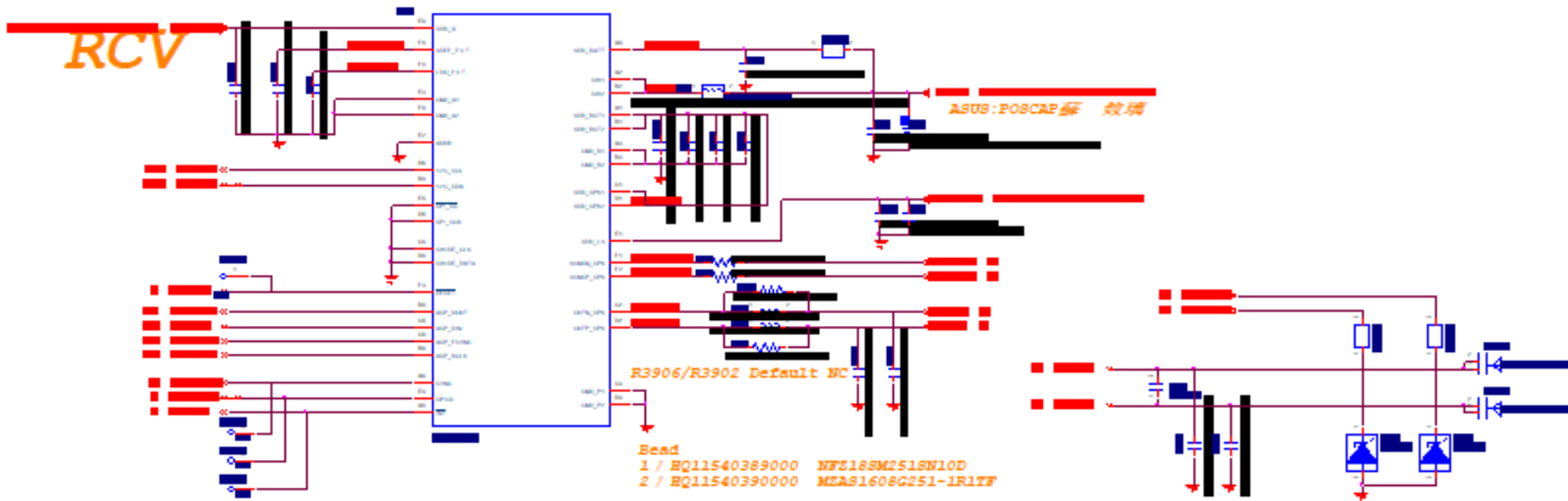
1. Check SMT status of RFB ANT2901, ANT2905 and the contact to MIC3 SB.
2. Confirm the defect is with MIC3 SB or MB.
3. Defect is with MIC3 SB, check SMT status of Flash LED.
4. Defect is with MB, check SMT status of RFB ANT2901, ANT2905, interposer, MB U1901.

Flash 迴MIC 票



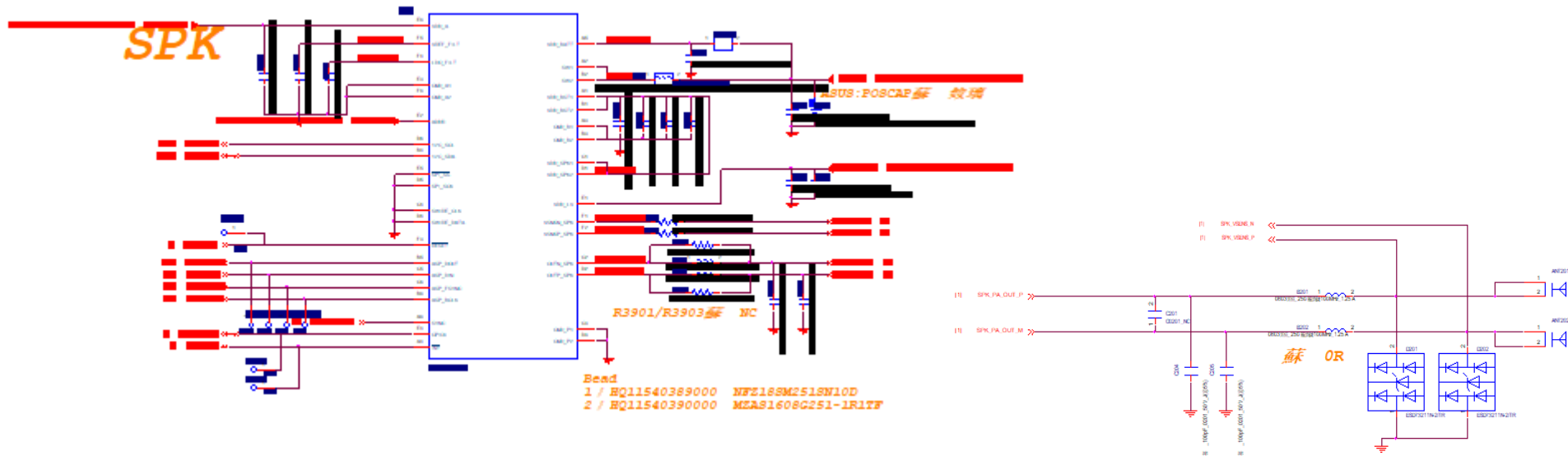
Receiver Fail

1. Check MB ANT3902, ANT3904 and the contact to Receiver is normal.
2. Check impedance of Receiver is not short.
3. Confirm the defect is with Receiver or MB.
4. Defect is with MB, check SMT status of MB ANT3902, ANT3904, and U3902 is normal.



Speaker Fail

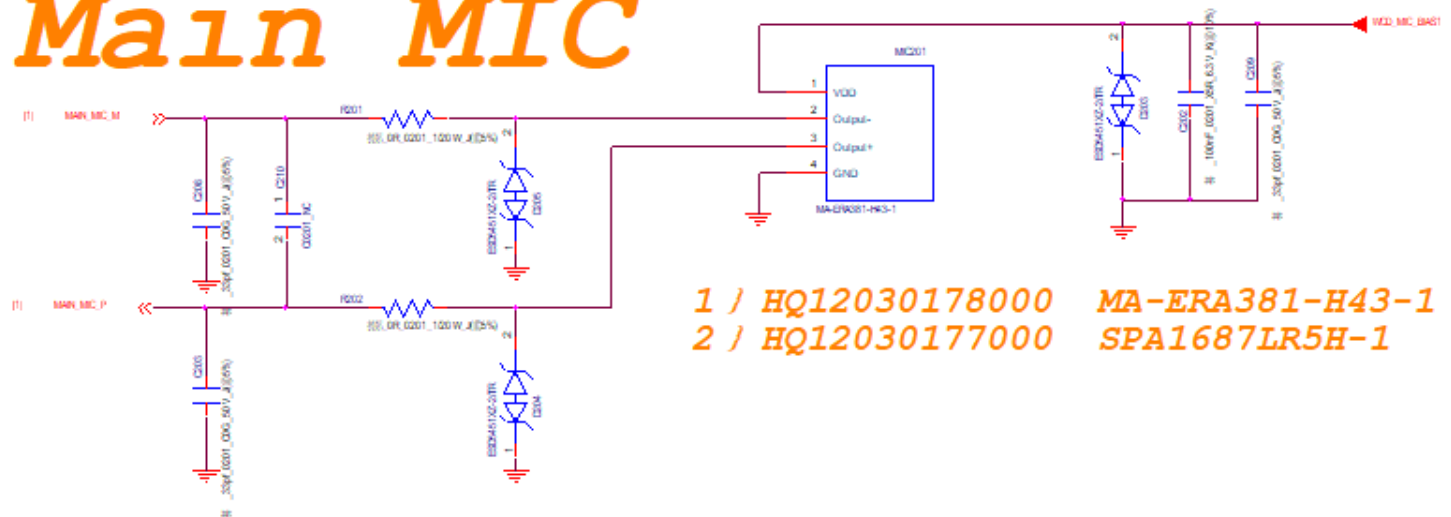
1. Check USB SB ANT201, ANT202 and the contact to Speaker is normal.
2. Check impedance of Speaker is not short.
3. Confirm the defect is with Speaker, IO Link FPC, USB SB or MB.
4. Defect is with USB SB, check SMT status of USB SB ANT201, ANT202, B201, B202, C201, C204, C205, D201, D202, J101 is normal.
5. Defect is with MB, check SMT status of MB J3701 and U3901 is normal.



Main Mic Fail

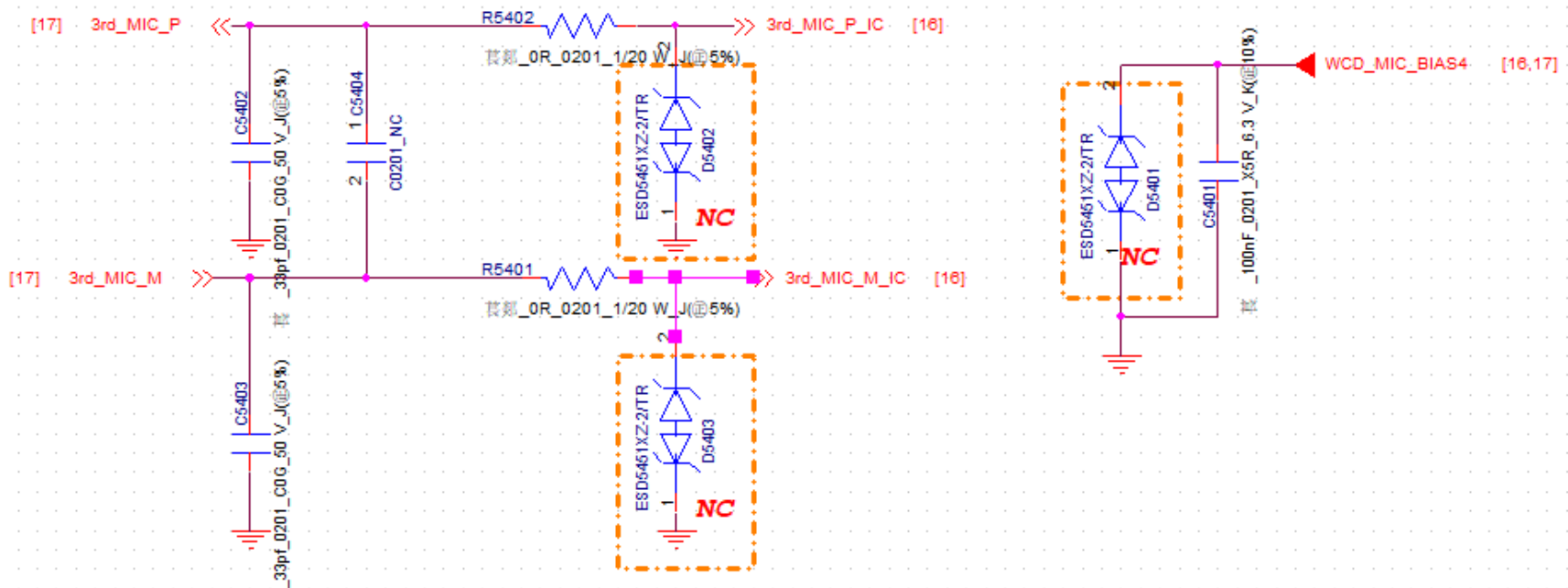
1. Confirm the defect is with IO Link FPC, USB SB or MB.
2. Defect is with USB SB, check SMT status of USB SB J101, C202, C209, D203, C208, C210, R201, D205, C203, R202, D204 MIC201 is normal.
3. Defect is with MB, check SMT status of MB U2901 is normal.

Main MIC



3rd Mic Fail

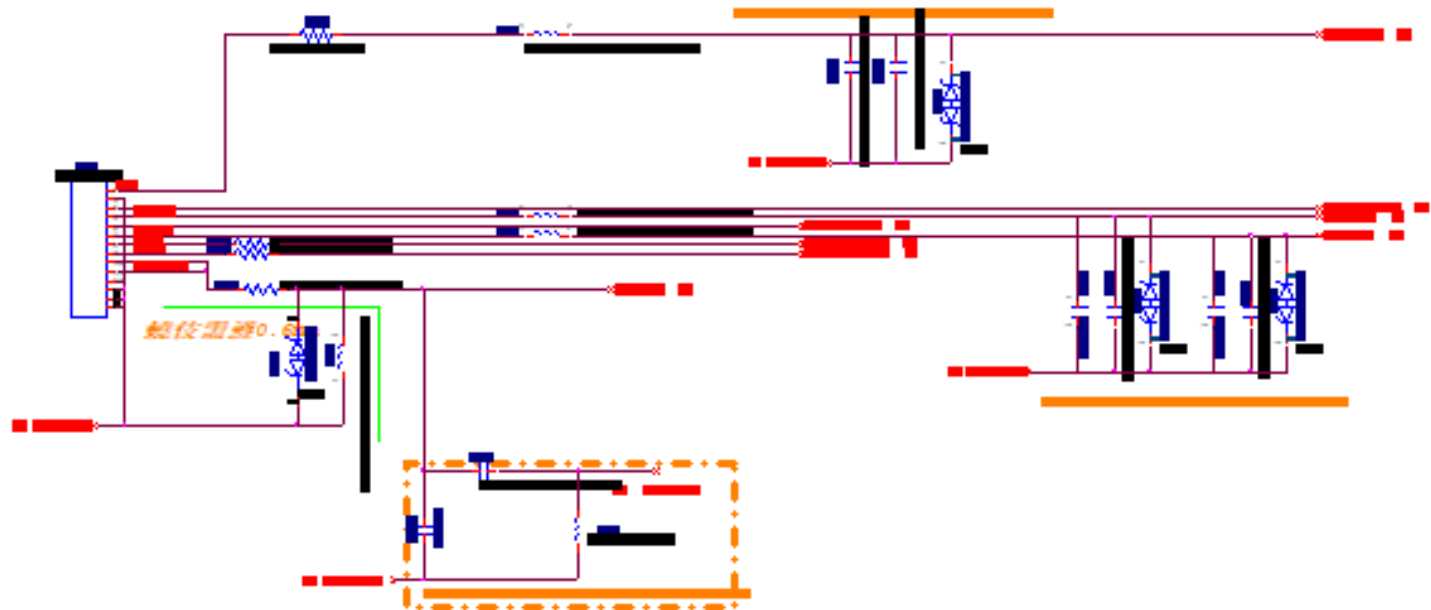
1. Check SMT status of RFB ANT2902, ANT2903, ANT2904, ANT2906 and the contact to MIC3 SB.
2. Confirm the defect is with MIC3 SB or MB.
3. Defect is with MIC3 SB, check SMT status of MIC3.
4. Defect is with MB, check SMT status of RFB ANT2902, ANT2903, ANT2904, ANT2906, C5402, C5404, D5402, R5402, C5401, D5401, C5403, D5403, R5401, interposer, MB U2901.



Headset Fail

1. Confirm the defect is with Headset FPC or MB.
2. Defect is with MB, check SMT status of RFB J5702, R5716, B5703, C5711, C5712, D5705, B5704, B5706, R5718, R5717, R5705, C5718, C5719, D5703, C5720, C5721, D5704, D5701, B5701, C5714, C5715, L5701, interposer, U2901 is normal.

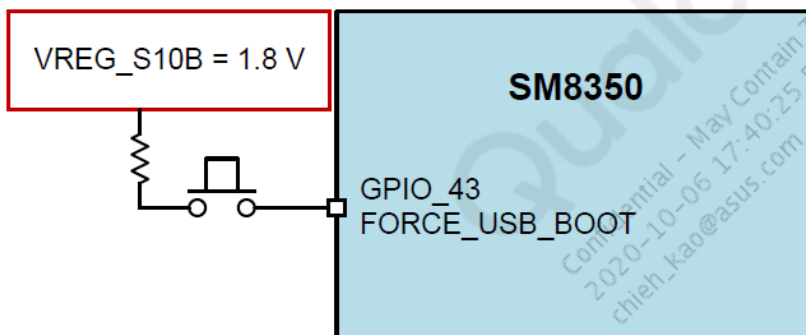
3. 5mm HEADSET



How to boot into 9008 mode

FORCE_USB_BOOT

- During development or factory production, a boot from the primary USB 3.1 port can be forced by using GPIO_43.
- FORCE_USB_BOOT (GPIO_43) always takes precedence, regardless of the state of the BOOT_CONFIG GPIOs or FAST_BOOT_SEL fuses.
- FORCE_USB_BOOT is checked first during the boot device detection, prior to BOOT_CONFIG GPIO states.
 - GPIO_43 = 1 forces the SM8350 device to boot from the primary USB 3.1 port.



Short the following two points

S10B_1P8 TP3840.1
BOOT TP3841.1

